

Advanced Attention Mechanisms - I

Large Language Models: Introduction and Recent Advances

ELL881 · AIL821



Sourish Dasgupta

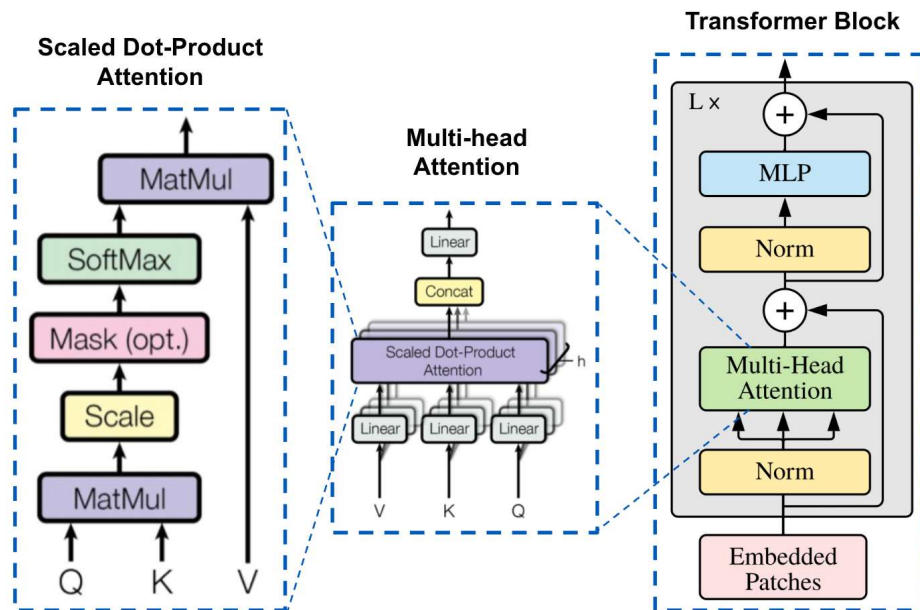
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Year: 2017, NeurIPS



Self Attention



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$\begin{matrix} \text{Q} & \times & \text{K}^T \\ (6, 4096) & & (4096, 6) \end{matrix} = \frac{\quad}{\sqrt{4096}}$$

	THE	CAT	IS	ON	A	CHAIR
THE	0.268	0.119	0.134	0.148	0.179	0.152
CAT	0.124	0.278	0.201	0.128	0.154	0.115
IS	0.147	0.132	0.262	0.097	0.218	0.145
ON	0.210	0.128	0.206	0.212	0.119	0.125
A	0.146	0.158	0.152	0.143	0.227	0.174
CHAIR	0.195	0.114	0.203	0.103	0.157	0.229

(6, 6)



Year: 2017, NeurIPS



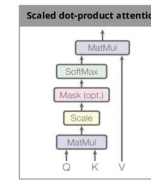
Causal (Forward Masked) Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



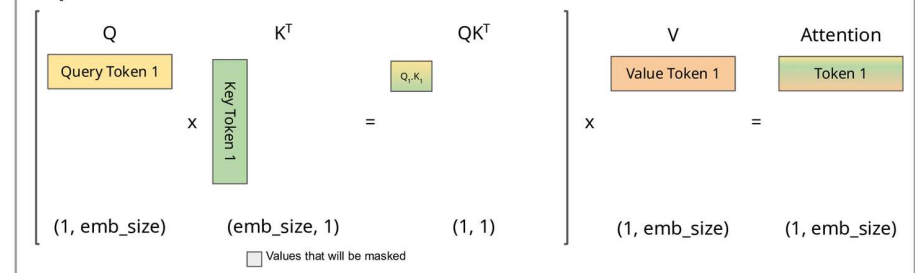
	THE	CAT	IS	ON	A	CHAIR
THE	0.268	---	---	---	---	---
CAT	0.124	0.278	---	---	---	---
IS	0.147	0.132	0.262	---	---	---
ON	0.210	0.128	0.206	0.212	---	---
A	0.146	0.158	0.152	0.143	0.227	---
CHAIR	0.195	0.114	0.203	0.103	0.157	0.229

(6, 6)



Time: $O(N^2 \cdot d) + O(N^2) + O(N^2 \cdot d) = O(N^2 \cdot d)$
 Space: $O(3 \cdot N \cdot d) + O(N^2) + O(N \cdot d) = O(N^2 + N \cdot d)$
 $= (3 \times N \times d + N^2) \times \text{Size of a float}$

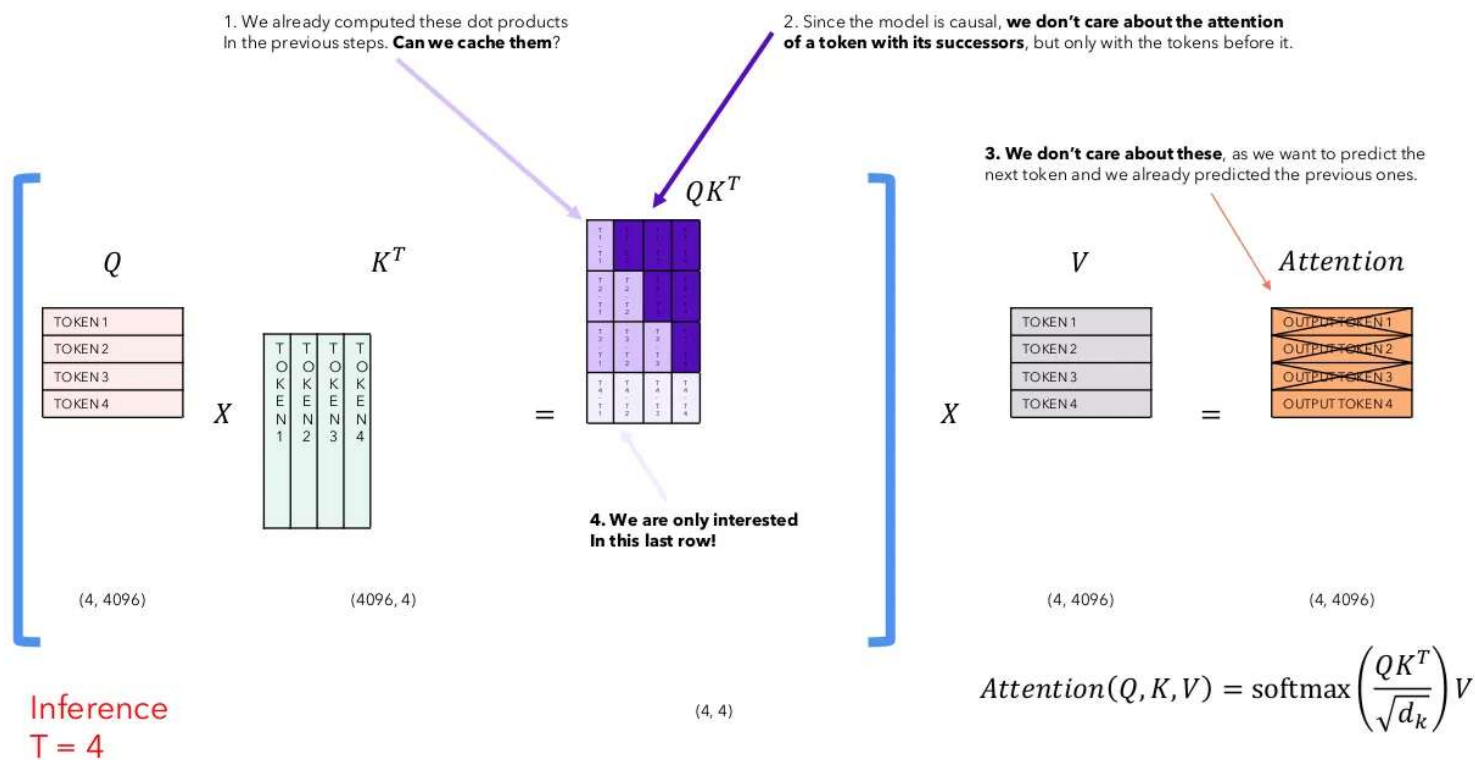
Step 1



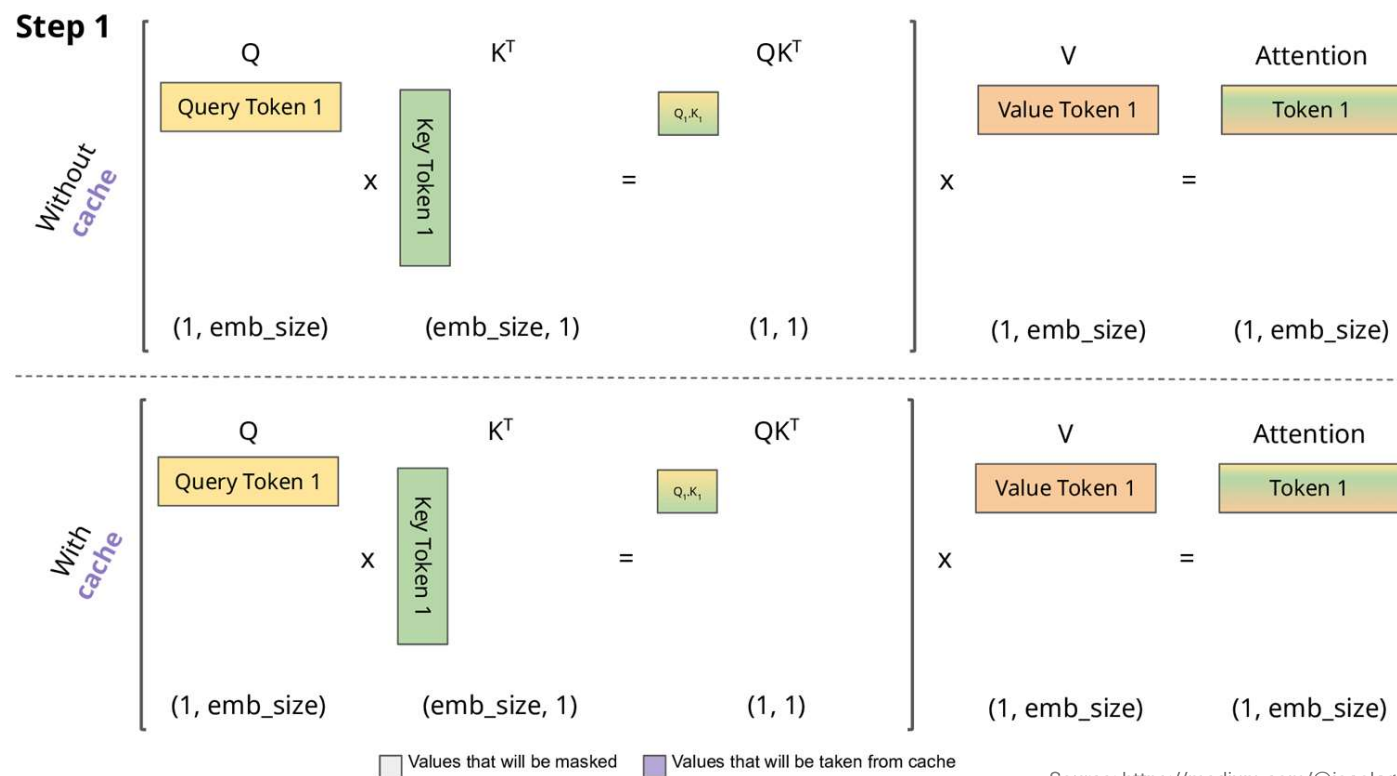
Zoom-in! (simplified without Scale and Softmax)



Why do we need to do better?



KV Cache based (Forward Masked) Attention



KV Cache Storage = $O(N \times d)$

$(2 \times N \times d) \times \text{Size of a float}$

vs.

$(3 \times N \times d + N^2) \times \text{Size of a float}$



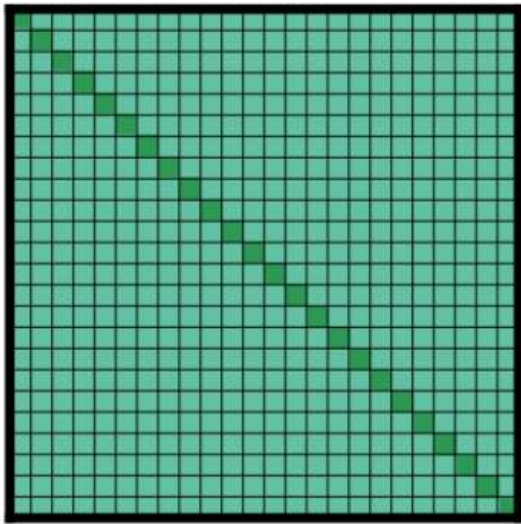
Source: <https://medium.com/@joaolages/kv-caching-explained-276520203249>



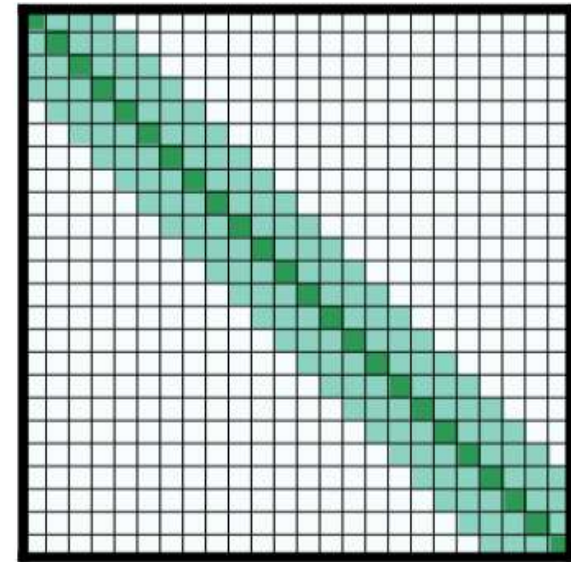
Year: 2020, Arxiv



Sliding Window Attention



(a) Full n^2 attention



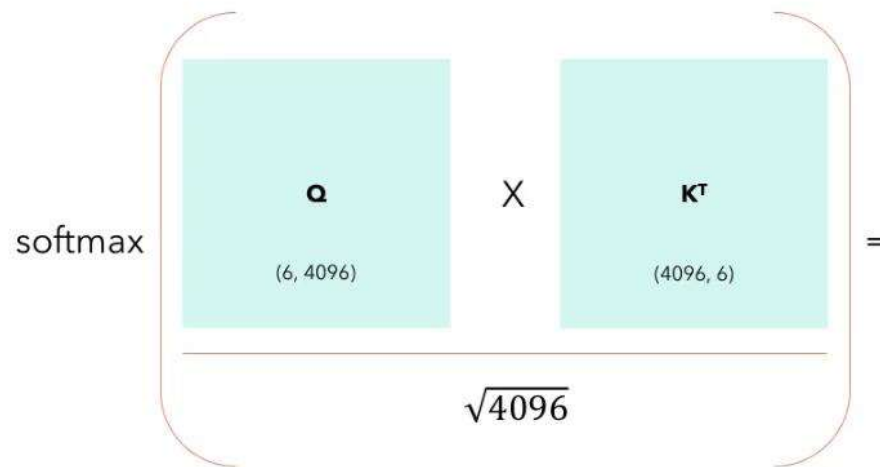
(b) Sliding window attention



Year: 2020, Arxiv



Sliding Window Attention



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

	THE	CAT	IS	ON	A	CHAIR
THE	1.0	0	0	0	0	0
CAT	0.461	0.538	0	0	0	0
IS	0.3219	0.317	0.361	0	0	0
ON	0	0.316	0.341	0.343	0	0
A	0	0	0.326	0.323	0.351	0
CHAIR	0	0	0	0.313	0.331	0.356

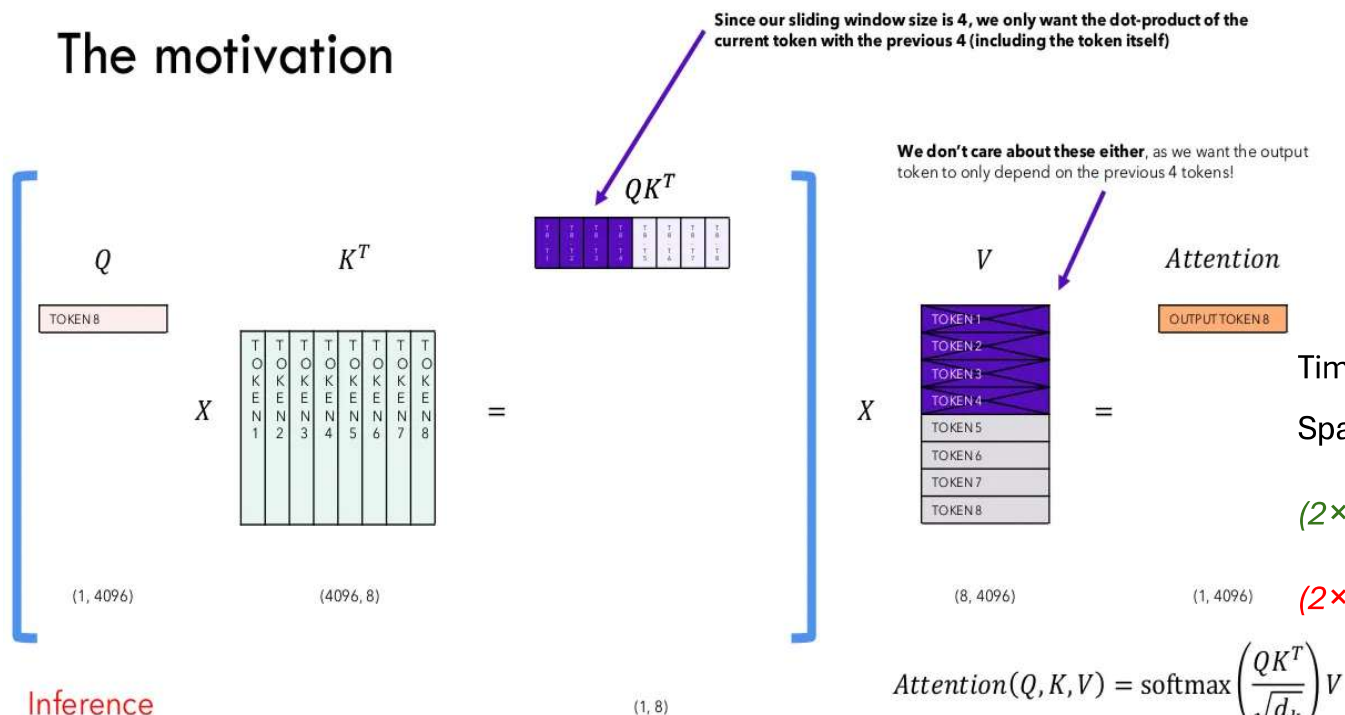


Year: 2020, Arxiv



What happens to the KV Cache?

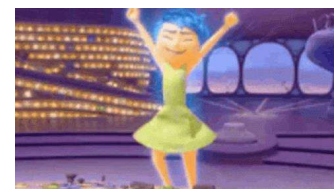
The motivation



Time: $O(N \cdot w \cdot d) + O(N \cdot w) + O(N \cdot w \cdot d) = O(N \cdot w \cdot d)$

Space: $O(2 \cdot N \cdot d) + O(N \cdot w) = O(N \cdot w + N \cdot d)$

$(2 \cdot \underline{w} \cdot d) \times \text{Size of a float}$
vs.
 $(2 \cdot N \cdot d) \times \text{Size of a float}$



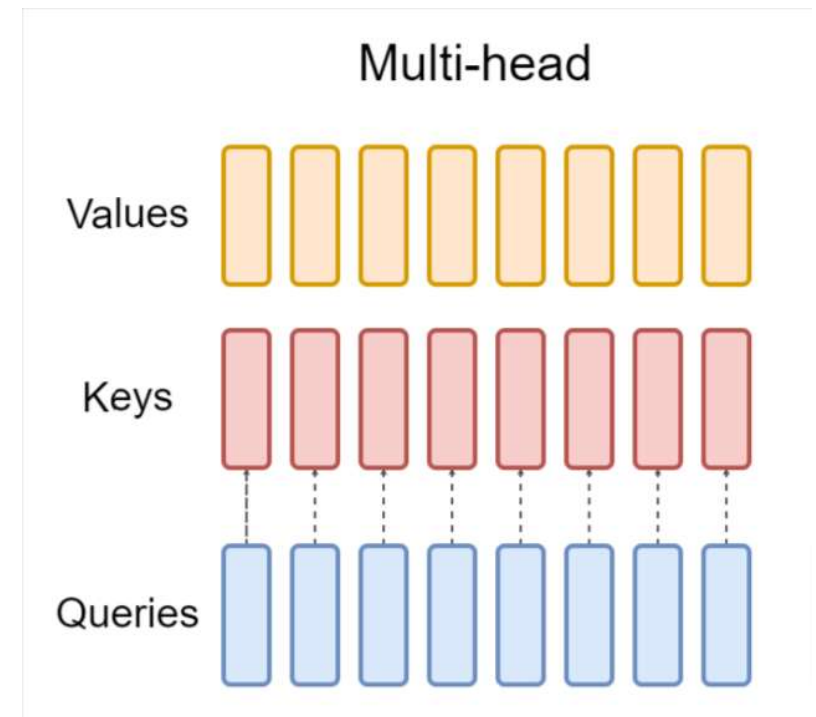
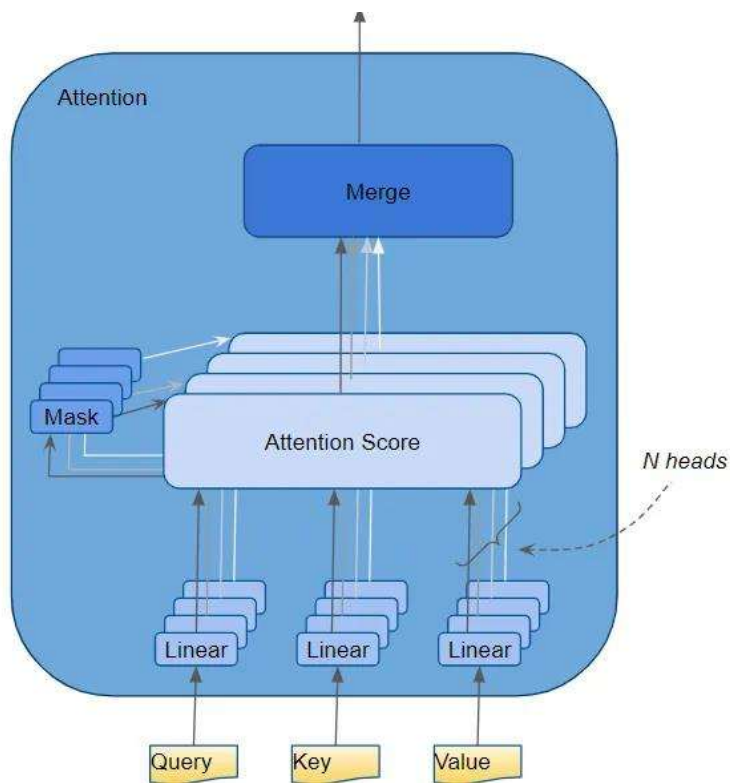
Inference
 $T = 4$



Going back to year: 2017, NeurIPS



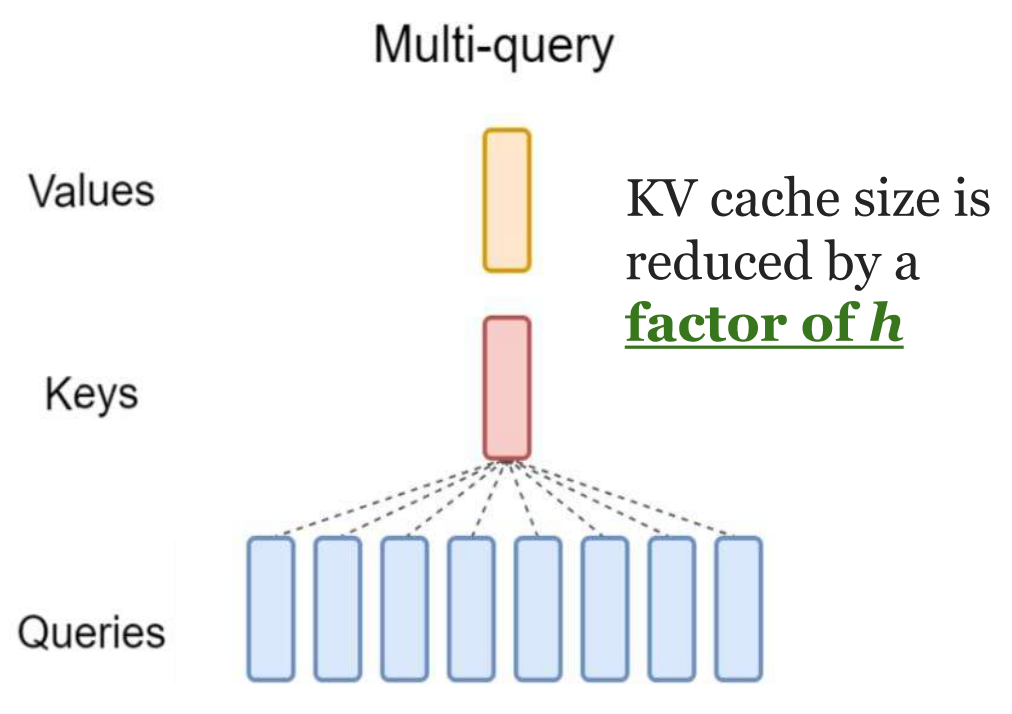
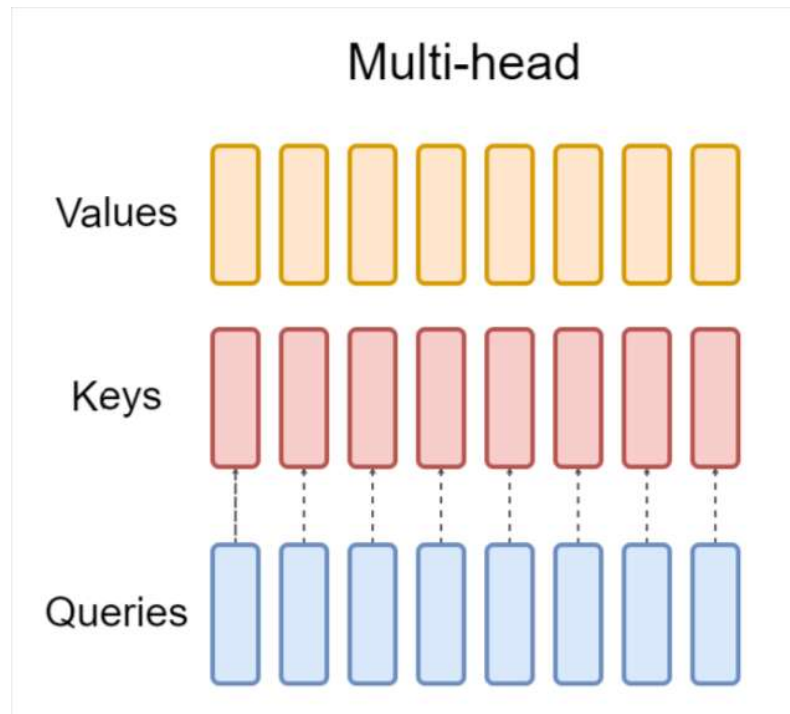
Multi-Head Self Attention



Year: 2019, Arxiv



Multi-Query Attention (MQA)



Do we lose out on something?

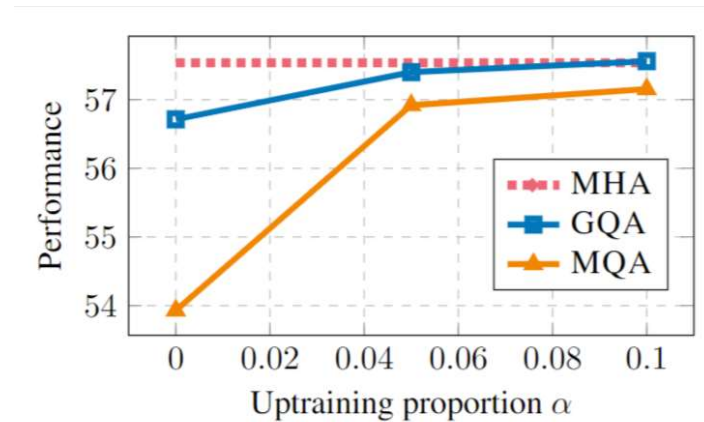
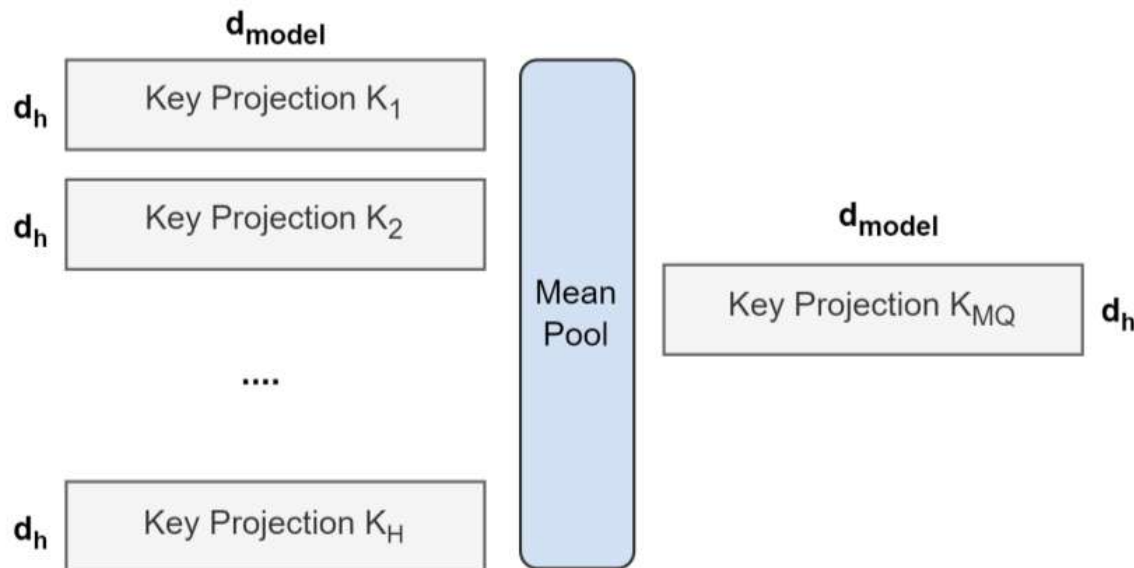
- Decline in performance quality
- Training instability



Year: 2023; ICLR



Uptraining: Converting MHA to MQA



What can still go wrong?

- Decline in performance quality
- ~~Training instability~~

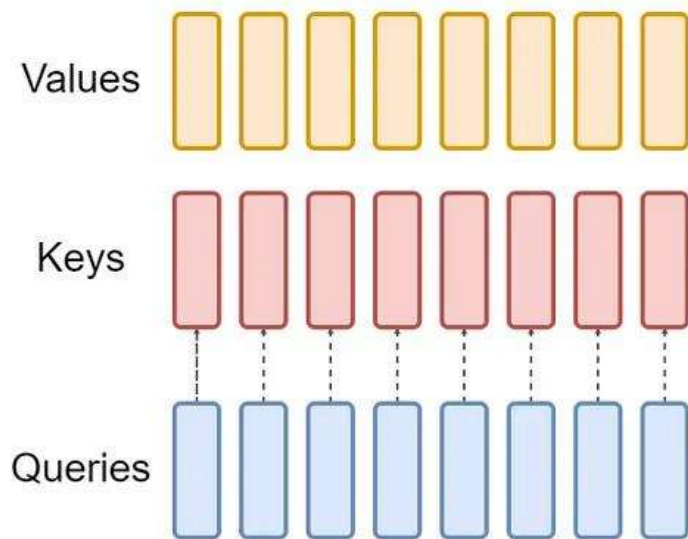


Year: 2023; EMNLP

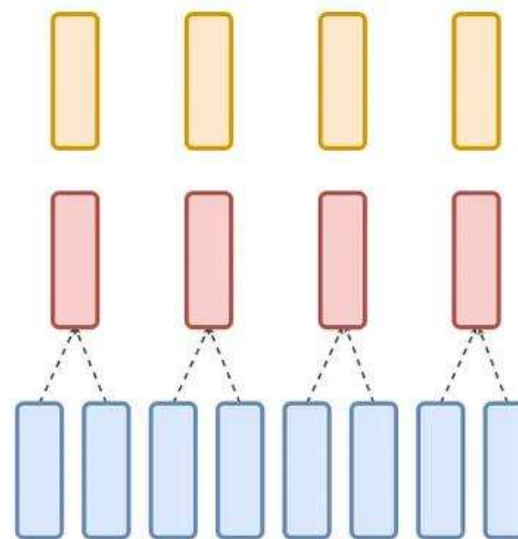


Grouped Query Attention

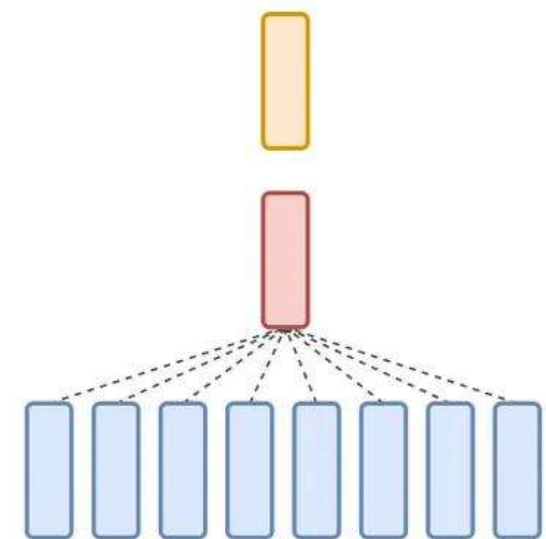
Multi-head



Grouped-query

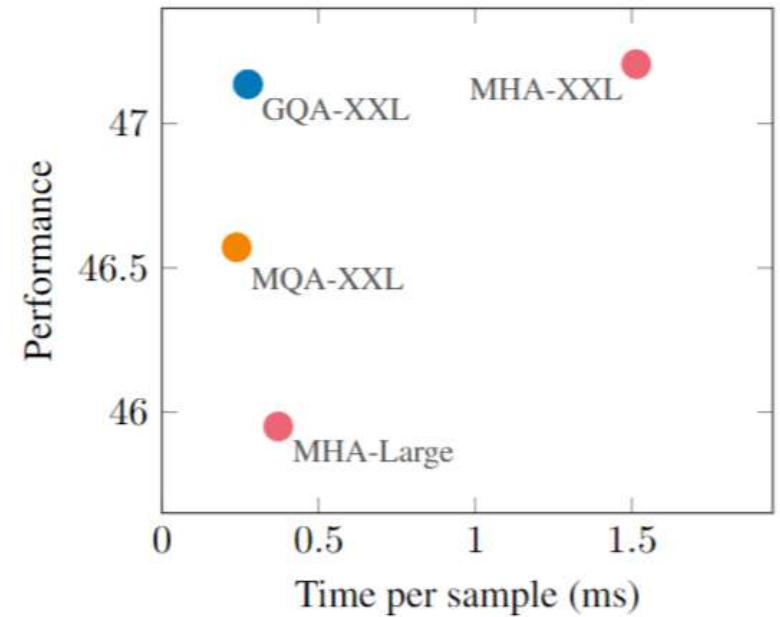
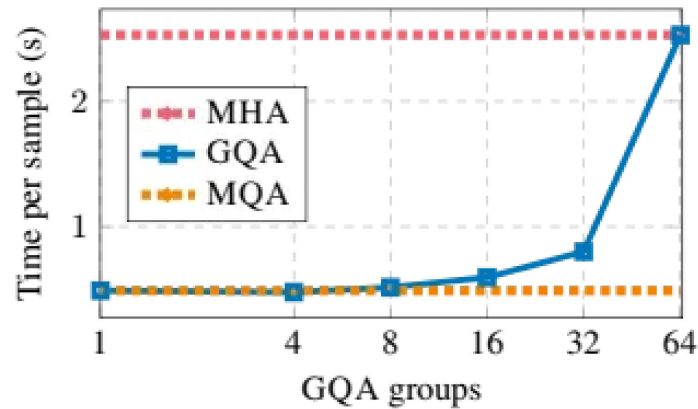


Multi-query



What did we gain?

Model	T_{infer}	Average	CNN	arXiv	PubMed	MediaSum	MultiNews	WMT	TriviaQA
	s		R_1	R_1	R_1	R_1	R_1	BLEU	F1
MHA-Large	0.37	46.0	42.9	44.6	46.2	35.5	46.6	27.7	78.2
MHA-XXL	1.51	47.2	43.8	45.6	47.5	36.4	46.9	28.4	81.9
MQA-XXL	0.24	46.6	43.0	45.0	46.9	36.1	46.5	28.5	81.3
GQA-8-XXL	0.28	47.1	43.5	45.4	47.7	36.3	47.2	28.4	81.6



So are we all set? Key Takeaways takeaways

- GQA/MQA Aim: *To reduce the need for storing a large amount of KV cache*
- LLM server can handle more requests, larger batch sizes and increased throughput
- *Cannot significantly reduce the computational load*
- *Quality degradation remains*

